

SOLVAER III, Norway

WMO: 011210

Lat: 66.371N

Lon: 12.611E

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.0	21.8	7.0	7.9	22.7	9.6	9.0	23.8	43.5	31.8	40.0	29.0	27.3	120	0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	70.1	59.6	66.5	57.3	63.6	56.0	60.9	68.0	58.7	64.3	57.1	61.9	12.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	70.8	63.1	56.0	66.8	61.4	54.5	63.3	59.7	27.0	68.4	25.6	64.7	24.5	61.9	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.8	33.4	29.9	DB	17.2	76.5	4.3	4.1	14.1	79.5	11.6	81.9	9.1	84.2	6.0	87.2	(4)
(5)			WB	14.7	63.5	3.4	3.2	12.3	65.8	10.3	67.7	8.3	69.5	5.8	71.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	33.3	33.2	34.8	40.3	46.5	50.8	56.0	56.4	52.6	45.4	39.3	36.0
	DBStd	9.80	5.61	5.86	4.84	4.76	5.55	4.92	5.20	4.20	4.18	4.66	5.31	5.48
	HDD50	2867	519	470	470	295	138	44	2	0	19	155	322	433
	HDD65	7749	984	890	935	740	573	425	287	267	371	608	771	898
	CDD50	602	0	0	0	5	30	69	188	200	98	12	0	0
	CDD65	10	0	0	0	0	0	1	7	2	0	0	0	0
	CDH74	38	0	0	0	0	1	2	32	3	0	0	0	0
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0
Wind	WSAvg	15.4	19.7	19.2	17.0	13.8	13.2	13.1	11.8	11.4	13.0	16.4	17.3	18.7
Precipitation	PrecAvg	67.6	6.4	5.9	5.8	4.4	3.9	3.8	3.9	4.5	7.3	7.8	6.3	7.6
	PrecMax	90.6	12.3	12.0	11.5	9.4	6.2	8.2	9.6	11.5	19.3	16.5	13.0	14.6
	PrecMin	49.7	0.0	1.2	1.1	1.0	1.2	0.2	1.0	0.3	1.0	1.7	0.6	0.2
	PrecStd	11.5	3.3	2.8	2.6	2.2	1.5	2.2	2.6	2.5	4.3	4.0	3.7	3.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	44.6	46.9	59.6	67.3	69.3	78.1	72.9	65.7	59.4	52.2	47.2
		MCWB	42.7	41.4	40.0	47.9	53.8	56.9	64.5	61.9	56.1	51.0	47.9	43.4
	2%	DB	43.8	43.4	44.1	54.9	62.1	65.0	72.5	68.6	63.0	56.3	49.5	45.9
		MCWB	40.2	40.6	40.7	45.8	51.5	54.8	61.8	59.1	54.4	49.1	45.7	43.4
	5%	DB	42.3	42.3	43.2	48.9	58.2	60.9	67.2	65.6	60.6	53.8	47.7	44.7
		MCWB	39.1	39.6	40.0	42.6	50.1	53.1	59.0	57.4	52.9	48.2	43.8	41.9
	10%	DB	40.5	40.9	41.8	46.0	54.1	57.5	63.0	63.1	58.7	51.3	46.1	43.2
		MCWB	37.5	38.0	38.5	41.1	47.5	52.0	57.0	56.5	52.2	47.3	42.2	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	43.1	43.5	48.2	55.2	58.6	66.3	62.9	57.3	52.7	49.3	45.9
		MCDB	45.1	43.6	44.1	58.8	66.4	67.5	76.4	71.1	63.0	58.2	50.8	46.4
	2%	WB	41.5	41.9	42.2	46.0	52.5	55.7	62.2	60.0	55.6	50.6	46.4	44.4
		MCDB	42.6	42.9	43.6	53.6	60.2	63.2	70.5	66.5	60.2	54.4	48.6	45.7
	5%	WB	39.6	40.1	40.4	43.7	50.4	53.8	59.9	58.3	54.3	49.0	44.5	42.2
		MCDB	41.5	41.8	42.8	47.5	57.5	59.1	65.7	64.0	58.6	52.4	47.6	44.1
	10%	WB	37.8	38.1	38.6	42.3	48.1	52.5	57.9	56.9	53.3	47.6	42.8	40.4
		MCDB	40.3	40.3	41.5	45.0	53.0	56.9	62.1	61.8	57.3	51.1	45.6	43.0
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.3	5.7	6.6	7.6	6.4	7.4	7.5	6.3	5.1	5.0	5.3
		MCDBR	6.2	6.5	6.7	9.9	14.6	13.1	15.7	12.7	9.0	6.5	5.5	6.7
		MCWBR	6.1	5.9	5.8	5.4	6.8	5.9	7.0	5.8	4.7	4.4	4.7	6.4
	5% WB	MCDBR	6.1	6.5	7.0	8.7	13.5	11.1	14.2	12.0	8.1	5.7	5.8	6.6
		MCWBR	6.3	6.5	6.8	5.1	7.2	5.5	7.0	5.9	4.9	5.1	5.7	6.7
Clear-Sky Solar Irradiance	taub	0.369	0.300	0.327	0.350	0.350	0.346	0.355	0.342	0.331	0.306	0.348	0.366	
	taud	2.539	2.182	2.282	2.361	2.443	2.500	2.500	2.525	2.521	2.323	2.500	2.489	
	Ebn at Noon	62	171	231	257	270	274	268	260	234	171	58	10	
	Edh at Noon	6	16	25	30	30	29	28	25	20	14	5	2	
All-Sky Solar Radiation	RadAvg	19	130	503	1012	1422	1598	1436	1088	588	221	39	2	
	RadStd	2	24	75	104	99	207	155	116	76	44	5	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	-167	-239	N/A	N/A

Nomenclature: See separate page